What is a Design Brief?

This document is a guide to writing the Design Brief. In addition to this explanation, you should also look at the textbook Chapter 4, focusing on Stages 1-3 of the Design Report. Those are the elements of the design report that go into a design brief.

Fundamentally, a design brief is a document that a design team uses to invite design ideas that satisfy a need. It is what is handed out at the beginning of a hack-a-thon, or by a client looking for contractors to present alternatives. It is also the baby sister to the "Request for Proposals" (RFP), a document you will write next term. The goal of a design brief is to set a team off in the right direction.

A Design Brief is aimed at a responding design team (which might be yourselves) who want to provide a good solution to the problem defined. They need to know what makes a solution "good" and they need to understand the nature of the problem (or opportunity) being solved. The responding team needs to know the boundaries of what is allowable in the design work and the nature of what is "better" or "worse."They do not need to know what you would make (your solution)—because if you already have a solution, why are you asking?

A good design brief invites multiple approaches to an opportunity. One test we use regularly with briefs is the 5×5 test: can you brainstorm 5 completely different solutions in 5 minutes? If not, maybe the design space is overly restricted, or maybe your opportunity is not very interesting. If there is only one solution, you don't need a brief, and you never really had an opportunity to begin with.

This document covers four things specific to what you are working on:

1. The logic of a design brief
2. The role of the claim in a design brief
3. How the design brief sets up the requirements framework
4. How the design brief sets up future work.

1. Defining a "problem" as the logic of a design brief

Engineering is regularly called a "problem-solving" discipline. So, it should be no surprise that "problem" lies at the centre of a design brief. The trick is, you should understand "problem" to include

- problems: things that are broken, not working, or otherwise in need of change
- opportunities: things that are interesting, open up new possibilities, or otherwise improve on or build the good
- challenges: things that seem like they could happen, should happen, or would be massively cool if they did. Things that are beyond the bounds of what we expect.

The problem with "problem" is that it sounds negative. Sometimes we use the fake word: "probletunity" just to remind ourselves that we can work in multiple directions. As long as we are clear that "problems" could actually be "opportunities," we can move forward.

Defining the "problem" is the goal of the design brief. It does **not** solve the problem. The brief should demonstrate that the problem is worth solving. To do this, we need to have a clear understanding of the problem from multiple perspectives, both human and technical.The design brief balances these two pieces.

The human side of the design brief

The human side of the design brief involves three critical elements: stakeholders, interpretations, and values.

Stakeholders are entities (individuals and groups but also non-human things) that are impacted by and/or impact a work of engineering design, and thus also the framing of a design opportunity. To get at stakeholders' interests, we might watch them, ask them, research them, or test with them. We do not simply do what they tell us because stakeholders may not understand the opportunity fully, or may see the opportunity only from their perspective. That's their stake after all, and we shouldn't expect them to look beyond it. Our responsibility in framing an opportunity is to integrate multiple perspectives, some of which might actually be at odds with one another.

That's where **interpretation** comes in. We interpret *our* understanding of stakeholders' expectations (as opposed to what they want or say they need). This involves judgment and subjectivity—they're subjects and so are we, so that should be expected.

Finally, our **values** form an important part of design work. A design team needs to establish its own unique perspective on a design opportunity. We see an opportunity in a particular way because of our values, so it helps to be able to articulate them. But it isn't enough to just say you have a value. You also need to explain what you mean. So, if you value sustainability, how do you define it? (Hint: you can almost always use an established definition.) Are there common principles of sustainability that apply in all cases, or does the context matter? Many of your values can become measurable and quantifiable through a range of DfX principles. Perhaps your design brief is the first place where you can make those things matter by building them into what you are looking for in the design.

The technical side of the design brief

The design brief is a technical document that establishes the requirements framework for a work of design. This is the hardest part of the brief. We have already discussed the requirements framework in class and in studio, so look at those resources.

Reference designs form an essential part of the design brief. Reference designs do two major things for us:

1. They explore previous attempts to solve our opportunity (or a similar one). The weaknesses of those attempts help identify critical areas that need to be resolved for a design that resolves the brief to be satisfactory.
2. Reference designs may show us features we want. These features do not even need to be for the same kind of device. For example, I might be setting up the stakeholder expectations for a new design of a super backpack, but I see an innovative handle on an umbrella that allows operation with only one hand. Then, I can add into my goal for my super backpack that it be operable with only one hand and offer the umbrella as a reference (note: you do not request the actual handle of the umbrella be included in the design, just show one way of enabling one-handed operation as a source of inspiration for the framing of the opportunity).

In this way, reference designs guide designers to combine ideas in new ways. Admittedly, they represent a danger as well: they could 'anchor' a team to a particular idea because of the reference. That's a careful balance we like to strike in the design brief, between giving some guidance and avoiding overly narrowing the range of possibility.

2. The role of the claim (and argument) in the design brief

A design brief is not an essay that has a thesis. It is a document that makes three main types of claim, all of them interconnected, and each of which is essential to the success of the brief. Lower-level claims support higher-level claims to create the overall argument of the design brief.

1. The claim that the stakeholder interpretation is legitimate. Admittedly, you do not come out and say, "This interpretation is legitimate." However, you do make claims about the stakeholder that need to be well justified by evidence from research and observation as noted above.

2. Each part of the breakdown of the need into goals and objectives is a claim. You are claiming that each goal is necessary to meet the need, and that each objective is a specific, measurable target that shows you have met a goal. Each should be justified and connected to a stakeholder expectation.

3. Each objective is a claim. In putting forward a objective, you are *claiming* that the metric will measure what you want it to measure, and allow someone to make a judgment about that aspect of a design. Each one needs to be defended with a suitable combination of research and reasoning.

3. Requirements framework makes the design brief

While all of the components of the brief are important, the quality and usability of the brief are most correlated with how well you define the stakeholder expectations with the need, goals and objectives. As noted above, several types of claims appear in the needs, goals, and objectives. As the starting point of the requirements framework (which also includes requirements and evaluation criteria at a later stage of the process), the need, goals, and objectives set the stage for complete framing of an engineering design opportunity.

Table 1 shows a sample set of objectives associated with one goal in a project that focused on reducing repetitive strain injuries experienced by maintenance staff who mopped floors and had to wring out mops many times each day. Notice that the goal is quite broad, whereas the objectives are specific and measurable. The objectives should be justified by relevant research; in this case, there is a Military Standard (MILSTD) that is relevant for Objectives 1.1 and 1.3.[1] Other types of objectives will rely on other kinds to research. The key here is that each objective is clear, with measurable targets that are properly justified using credible evidence and suitable logical reasoning.

You do not want to make arbitrary constraints. The constraint related to Objective 1.2 in the table has that problem: why 10% reduction? What if my design achieved 9.5% reduction? Or, as you can see from the design discussed on p. 105 in the textbook, what if we let go of that constraint altogether but still met the higher objective? Although it is possible to justify some constraints just using logical reasoning, this should be a last resort. It is much better to use good research to justify constraints.

[1] Pro-tip: MILSTD1472D happens to be the standard that deals with ergonomic design for the human body. It may be useful to some of you even in this brief. It's available online.

Table 1. A goal and three associated objectives for reducing repetitive strain injuries when mopping.

	Description	Metric	Justification
Goal 1	Reduce repetitive strain injury (RSI) related to mopping		
Objective 1.1	Reduce the force required to wring the mop by the user's arms to be as low as possible, and at least less than 100N	Arm force applied to wring the mop (N)	Reduce the force to below the current baseline. To reduce RSI the applied force must be less than 100N, as per MILSTD-1472D.
Objective 1.2	Reduce the number of wringing cycles to be as low as possible and less than 86 per day	Mop wringing cycles per day	The current baseline is an estimated 96 cycles per day, and the primary stakeholder would like at least a 10% reduction in their required work
Objective 1.3	Allow users to wring the mop without needing to bend more than 0.5 m below the recommended working height (Appendix B)	Required bending distance below the recommended working height (m)	Excessive bending can lead to additional strain. As per MILSTD-1472D; working height methods outlined in Appendix B

4. Set up the future

The design brief is a starting point, not an end point. If your brief works well, it allows teams to create requirements and evaluation criteria as part of their design process that support them in generating a range of ideas and assessing them in meaningful ways. The design team may push back on some of your framing, either because it's irrelevant to their design or because they have a new understanding of the stakeholder expectations.

The design brief is not set in stone. As you learn more, your interpretation may change, and you may need to modify aspects of the framing. New design teams will bring their own interpretations and perspectives to the brief and may make their own modifications. Normally, changes to the need, goals and objectives would require approval from (or at least consultation with) the major stakeholders. In some cases, especially in a course context like Praxis, we'll need to use a modified approach that may involve argument rather that stakeholder approval. Still, a design brief is a meaningful guide that will allow teams to move forward and develop an initial set of requirements and evaluation criteria that will define a design space where there are many viable possible concepts that satisfy the need.